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Module Objectives

Discrete Logarithm Problem

- Video: Discrete Logarithm
2 min
- Video: Primitive Root of a Prime Integer
3 min
- Video: Discrete Logarithm Problem
1 min
- Quiz: Discrete Logarithm and Primitive Root
Submitted

Diffie-Hellman Key Exchange

Congratulations! You passed!

Grade
received 100%

Latest Submission
Grade 100%

To pass 80% or higher

Go to next item

Review Learning Objectives

1. What is the value of the discrete logarithm of 3 base 2 mod 5?

4 / 4 points

3

Due Apr 28, 11:59 PM IST

Attempts 3 every 8 hours

Try again

Correct
This was also solved in the Primitive Root of a Prime Number lecture.
To Pass 80% or higher

Your grade
100%
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2. What is the value of the discrete logarithm of 4 base 5 mod 7?

4 / 4 points

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2

Correct
See Discrete Logarithm lecture.

3. Identify the primitive roots of the prime number 3. Select all that applies.

4 / 4 points

2

Correct
See Primitive Root of a Prime Integer lecture and compute the entries for the table.

0

1

4. Identify the primitive roots of the prime number 7. Select all that applies.

6 / 6 points

2

1

5

Correct
See Primitive Root of a Prime Integer lecture and compute the entries for the table.

3

Correct
See Primitive Root of a Prime Integer lecture and compute the entries for the table.

6

4

5. Which of the following statements are true?

4 / 4 points

The discrete logarithm mod p always exist and is unique if p is a prime number.

Using the primitive roots of a prime modulus p yields the maximum p-1 possible outcome values for the discrete logarithm, which is desired for cryptography.

Correct
If the modulus n is not a prime integer, then this only yields phi(n) possible outcome values where phi(n) is the Euler totient function and is less than n-1.

Given a large modulus n, the exponential operation mod n is computationally difficult.

Given a large modulus n, the discrete logarithm problem is computationally difficult.

Correct
See Discrete Logarithm Problem lecture